

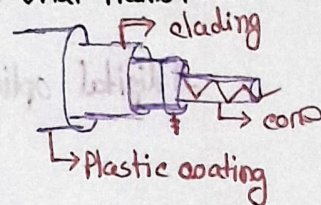
Chapter-1 (slide)

Communication →

Communication may be broadly defined as the transformation of information from one point to another.

Optical fiber →

Optical fiber is a thin, flexible strand of glass or plastic that transmits light signals over long distance with minimal loss.



Communication system →

A communication system is a setup that enables the transmission of information from one place to another, using various mediums.

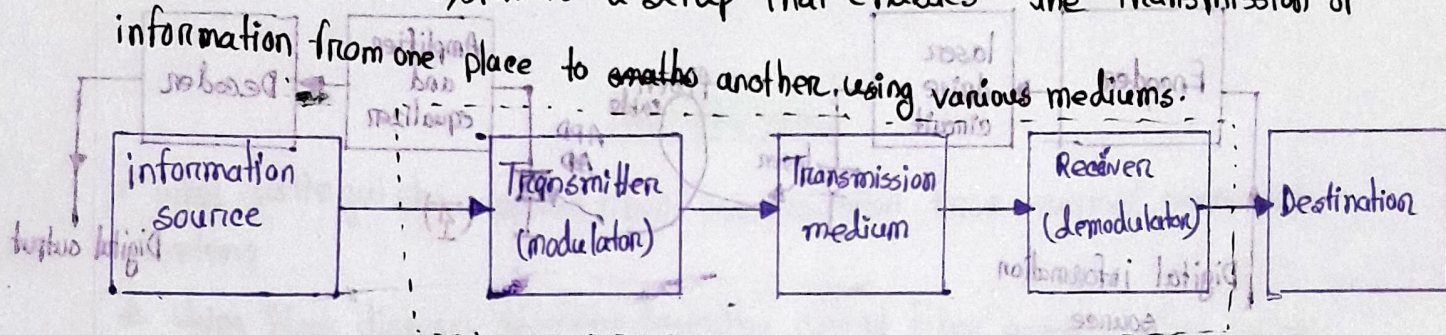
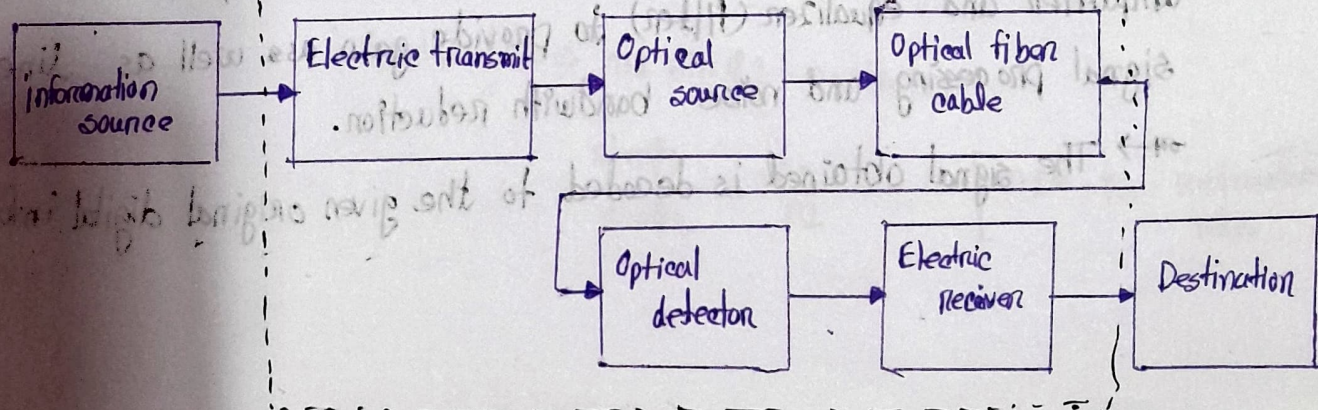


fig- Communication system

Optical fiber communication system →

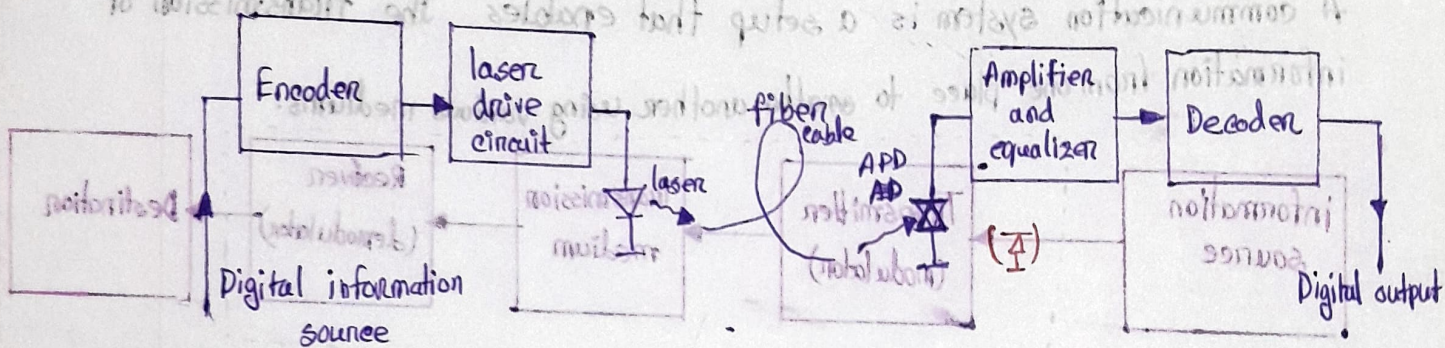
An optical fiber communication system is a type of communication system that transmits information using light signal through optical fiber.



Why analog optical fiber communication limited to short distance?

Analog modulation with optical fiber communication require higher signal-to-noise ratio at the receiver ~~is~~ than digital modulation, which make it less efficient. for this reason analog optical fiber communication links are generally limited to shorter distance and lower bandwidth.

A digital optical fiber link using semiconductor laser source and APD detector →



→ Input digital signal from information source encoded for optical transmission.

→ The laser drive circuit directly modulates the intensity of semiconductor laser with encoded digital signal.

→ A digital optical signal is launched into optical fiber.

→ An avalanche photodiode (APD) detector followed by a front-end amplifier and equalizer (filter) to provide gain as well as linear signal processing and noise bandwidth reduction.

→ The signal obtained is decoded to the given original digital information.

Advantage of optical fiber communication →

- Enormous potential band-width
- Small size and weight
- Electrical isolation
- Immunity to interference and crosstalk
- Signal security
- Low transmission loss.
- Ruggedness and flexibility
- System reliability and ease of maintenance.
- Potential low cost.

Read details from slide

Previous year

What distinguishes optical fiber communication from normal communication system?

Using Block diagram describe optical fiber communication system.

What is optical fiber?

Advantage of optical fiber communication.

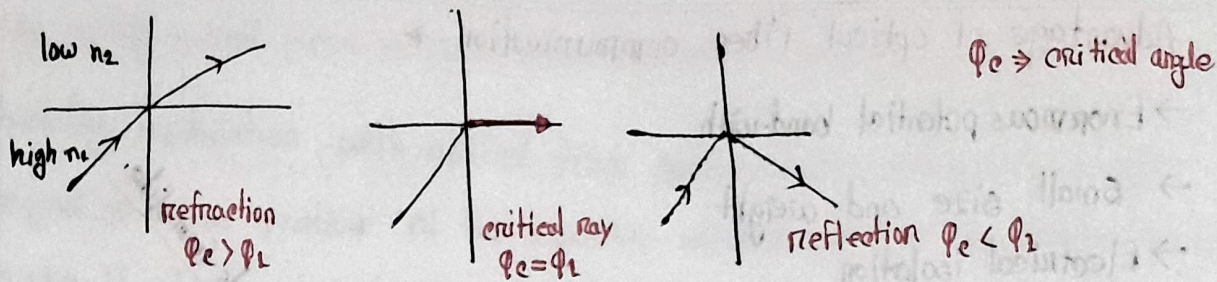
Chapter-2 (slide)

Angle of incidence Angle of incidence ϕ_1 , refraction ϕ_2 .

Snell's law $\Rightarrow n_1 \sin \phi_1 = n_2 \sin \phi_2$

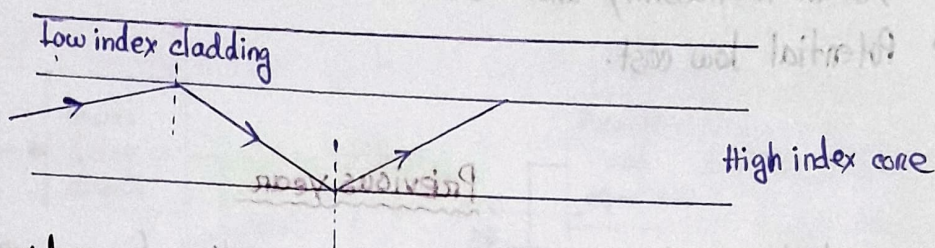
$$\therefore \frac{\sin \phi_1}{\sin \phi_2} = \frac{n_2}{n_1}$$

[n_1, n_2 = Refractive index]



critical angle $\phi_c = \sin^{-1} \frac{n_2}{n_1}$

Transmission of light ray in perfect optical fiber $\rightarrow$



At angle of incidence greater than critical angle the light is reflected back into the originating dielectric medium with high efficiency.

Acceptance angle $\theta_a \rightarrow$

The acceptance angle in the optical fiber is the maximum angle at which light can enter the fiber and still be guided through it by total internal reflection.

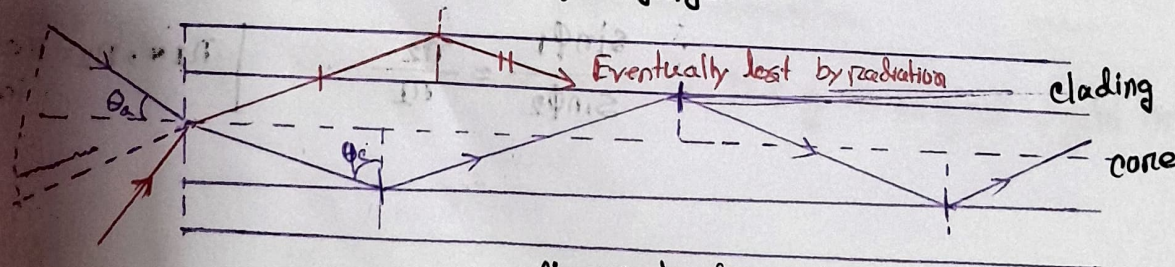
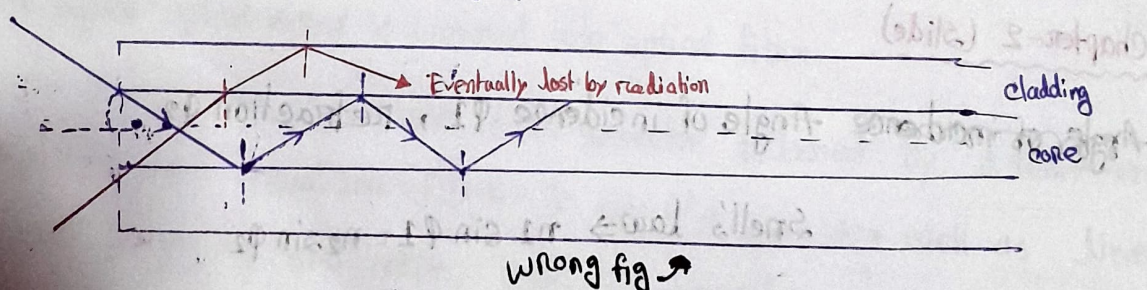


fig - Angle of Acceptance

Math for Chapter-2 →

Example 2.1 -

A silica optical fiber with a core diameter large enough to be considered by ray theory analysis has a core refractive index of 1.50 and cladding refractive index 1.47.

Determine - a) critical angle at core cladding interface.

b) the NA for the fiber.

c) Acceptance Angle in the air for the fiber.

Soln - a) Critical angle $\theta_c = \sin^{-1} \frac{n_2}{n_1}$

$$= \sin^{-1} \frac{1.47}{1.50}$$

$$= 78.5^\circ$$

core refractive index
 $n_1 = 1.50$

cladding refractive index
 $n_2 = 1.47$

b) Numerical aperture $(NA) = (n_1^2 - n_2^2)^{1/2}$

$$= (1.50^2 - 1.47^2)^{1/2}$$

$$= (2.25 - 2.16)^{1/2}$$

$$= 0.30$$

c) Angle of Acceptance $\theta_a = \sin^{-1} NA$

$$= \sin^{-1} 0.30$$

$$= 17.4^\circ$$

n_0 = Refractive index
in Air = 1

Example 2.2 -

A typical relative refractive index difference for an optical fiber design for long distance transmission is 1%. Estimate the NA and the solid acceptance angle in Air for the fiber where the core is 1.46. Further calculate the critical angle at the core-cladding interface within the fiber.

$$\begin{aligned} \text{NA} &= n_1 (2\Delta)^{1/2} \\ &= 1.46 (2 \times 0.01)^{1/2} \\ &= 0.21 \end{aligned}$$

Solid acceptance angle in air $\mathcal{J}$ -

$$\mathcal{J} \simeq \pi \theta_a^2 = \pi \sin^2 \theta_a$$

$$\begin{aligned} \mathcal{J} &\simeq \pi (\text{NA})^2 = \pi \times 0.04 \\ &= 0.13 \text{ rad} \end{aligned}$$

Relative refractive index difference $\Delta = \frac{n_1^2 - n_2^2}{n_1^2}$

$$= 1 - \frac{n_2}{n_1}$$

$$\therefore \frac{n_2}{n_1} = 1 - \Delta$$

$$= 1 - 0.01$$

$$= 0.99$$

critical Angle at core-cladding interface -

$$\theta_c = \sin^{-1} \frac{n_2}{n_1}$$

$$= \sin^{-1} 0.99$$

$$= 81.9^\circ$$

$$\text{NA} = n_0 \sin \theta_a$$

$$\text{NA} = (n_1^2 - n_2^2)^{1/2}$$

Relative Refractive index

$$\text{Difference } \Delta = \frac{n_1^2 - n_2^2}{2n_1^2}$$

$$= \frac{n_1 - n_2}{n_1} = 0.01$$

$$\therefore \text{NA} = n_1 (2\Delta)^{1/2}$$

[core-cladding interface]

Example 2.3 -

- An optical fiber in air has NA of 0.4. Compare the acceptance Angle for meridional rays with that for skew rays which change the direction by 100° at each reflection.

Soln -

$$\theta_a = \sin^{-1} NA$$

$$= \sin^{-1} 0.4$$

$$= 23.6^\circ$$

The skew ray change direction by 100° at each reflection, therefore $\gamma = 50^\circ$

The acceptance angle for skew ray is -

$$\theta_{as} = \sin^{-1} \left(\frac{NA}{\cos \gamma} \right) = \sin^{-1} \left(\frac{0.4}{\cos 50^\circ} \right) = 38.5^\circ$$

The acceptance Angle for skew ray is ~~greater~~ about 15° greater than the corresponding angle for meridional rays.

Chapter-3

Attenuation in optical fiber -

A number

Chapter - 3

Math for Chapter 3 -

Example 3.1-

When the mean optical power launched into 8 km length of fiber is 120 μ W, the mean optical power at the fiber output is 3 μ W.

- The overall signal attenuation or loss in decibels through the fiber assuming there are no connectors or splices.
- The signal attenuation per kilometer for fiber.
- The overall signal attenuation for a 10 km optical fiber link using the same fiber, with splices at 1 km intervals, each giving an attenuation of 0.1 dB.
- The numerical input/output power ratio in (c).

Soln-

a) Single attenuation =
$$10 \log_{10} \frac{P_1}{P_2}$$
$$= 10 \log_{10} \frac{120 \times 10^{-6}}{3 \times 10^{-6}}$$
$$= 10 \log_{10} 40$$
$$= 16.0 \text{ dB}$$

P_1 = input optical power
 P_2 = output optical power

b) Signal attenuation per kilometer length
$$\alpha_{\text{dB}} = 10 \log_{10} \frac{P_1}{P_2}$$
$$\Rightarrow \alpha_{\text{dB}} L = 16 \text{ dB}$$
$$\Rightarrow \alpha_{\text{dB}} = \frac{16}{8}$$
$$= 2 \text{ dB km}^{-1}$$

c) As $\alpha_{\text{dB}} = 2 \text{ dB km}^{-1}$, the loss for 10 km = 2×10
$$= 20 \text{ dB}$$

However there also 9 splices (at 1km intervals) each with an attenuation of 1dB. Therefore loss due to splice is 9dB.

$$\text{overall signal attenuation} = 20 + 9 \\ = 29 \text{ dB}$$

d) Input/output power ratio $\frac{P_1}{P_2} = 10^{(dB/10)}$
 $= 10^{(29/10)}$
 $= 794.3$

Example - 3.2 -

Silica has an estimated fictive temperature of 1400 K with an isothermal compressibility of $7 \times 10^{-11} \text{ m}^2 \text{ N}^{-1}$. The refractive index and photoelastic co-efficient for silica are 1.46 and 0.286 respectively. Determine the theoretical attenuation in dB per kilometer due to fundamental Rayleigh scattering in silica at optical wavelengths 0.63, 1.00 and 1.20 μm . Boltzmann's constant is $1.381 \times 10^{-23} \text{ J K}^{-1}$.

Soln -

Raylength scattering co-efficient

$$Y_R = \frac{8\pi^3 n^8 p^2 \beta_e k T}{3 \lambda^4}$$

Y_R = Raylength scattering co-efficient

λ = optical wavelength

n = refractive index

P = photoelastic coefficient

β_e = isothermal compressibility

k = Boltzmann constant

T = temperature

$$= \frac{248.15 \times 20.65 \times 6.082 \times 7 \times 10^{-11} \times 1.381 \times 10^{-23} \times 1400}{3 \times \lambda^4}$$

$$= \frac{1.895 \times 10^{-28}}{\lambda^4} \text{ m}^{-1}$$

At wavelength $0.63 \mu\text{m}$ -

$$\gamma_R = \frac{1.895 \times 10^{-28}}{0.158 \times 10^{-24}} = 1.199 \times 10^{-3} \text{ m}^{-1}$$

transmission loss factor for 1 kilometer of fiber

$$L_{\text{km}} = \exp(-\gamma_R L) = \exp(-1.199 \times 10^{-3} \times 10^3)$$

$\xrightarrow{\text{length}}$

$$= 0.301$$

Attenuation in dB per kilometer = $10 \log_{10} (1/L_{\text{km}})$

$$= 10 \log_{10} \left(\frac{1}{0.301} \right)$$

$$= 5.2 \text{ dB km}^{-1}$$

At wavelength of $10 \mu\text{m}$ -

$$\gamma_R = \frac{1.895 \times 10^{-28}}{10^{-24}} = 1.895 \times 10^{-4} \text{ m}^{-1}$$

loss factor $L_{\text{km}} = \exp(-\gamma_R L) = \exp(-1.895 \times 10^{-4} \times 10^3)$

$$= 0.827$$

Attenuation per in dB per kilometer = $10 \log_{10} (1/L_{\text{km}})$

$$= 0.8 \text{ dB km}^{-1}$$

At wavelength $1.30 \mu\text{m}$ -

$$\gamma_R = \frac{1.895 \times 10^{-28}}{2.856 \times 10^{-24}} = 6.64 \times 10^{-5} \text{ m}^{-1}$$

$$L_{km} = \frac{1}{\gamma_{RL}} (-\gamma_{RL}) = (-0.664 \times 10^4 \times 10^{-3}) = 0.936$$

$$\text{Attenuation per dB km}^{-1} = 10 \log_{10} (1/L_m) = 10 \log_{10} 1.069$$

$$= 0.3 \text{ dB km}^{-1}$$

Example - 3.3

A long single mode optical fiber has a attenuation of 0.5 dB km^{-1} when operating at a wavelength of $1.3 \mu\text{m}$. The fiber core diameter is $6 \mu\text{m}$ and laser source bandwidth is 600 MHz . Compare the threshold optical power for simulated Brillouin and Raman scattering within the fiber at the wavelength specified.

Simulated Brillouin Scattering (SBS)

Simulated Raman Scattering (SRS)

Soln - The threshold optical power for SBS -

d = fiber core diameter

λ = Operating wavelength

α_{dB} = fiber attenuation / km

V = Source of bandwidth

$$P_B = 4.4 \times 10^{-3} d^2 \lambda^2 \alpha_{dB} V \text{ watt}$$

$$= 4.4 \times 10^{-3} \times 6^2 \times 1.3^2 \times 0.5 \times 0.6$$

$$= 0.0803 \text{ Watt}$$

$$= 80.3 \text{ mW}$$

The threshold optical power for SRS -

$$P_R = 5.9 \times 10^{-2} d^2 \lambda^2 \alpha_{dB}$$

$$= 5.9 \times 10^{-2} \times 6^2 \times 0.5$$

$$= 1.38 \text{ W}$$

Brillouin threshold occurs at an optical power level 80 mW that is 17 larger than Raman threshold.

Example - 3.4

Two ^{step} index fibers exhibit the following parameters-

a) A multimode fiber with a core refractive index 1.500, a relative refractive index difference of 3% and an operating wavelength of 0.82 μm .

b) A 8 μm core diameter single mode fiber with a core refractive index 1.500 and relative refractive index difference of 0.3% and operating wavelength of 1.55 μm .
Estimate critical radius of curvature at which large bending loss occurs.

Soln -

a) Refractive index difference $\Delta = \frac{n_1^2 - n_2^2}{2n_1^2}$ $n_1 = \text{core}$
 $n_2 = \text{cladding}$
 $n_2^2 = n_1^2 - 2\Delta n_1^2$
 $= 2.250 - 0.06 \times 2.250$
 $= 2.175$

~~Q.1~~ $\Rightarrow$ Multimode fiber critical radius of curvature -

$$R_c \approx \frac{3n_1^2 \lambda}{4\pi(n_1^2 - n_2^2)^{3/2}} = \frac{3 \times 2.250 \times 0.82 \times 10^{-6}}{4\pi \times (0.135)^{3/2}}$$

$$= 9 \mu\text{m}.$$

b) Refractive index difference $\Delta = \frac{n_1^2 - n_2^2}{2n_1^2}$

$$\therefore n_1^2 = 2.237$$

The cut-off wavelength for single mode fiber

$$\lambda_c = \frac{2\pi a n_1 (2\Delta)^{1/2}}{2.405}$$

λ_c ← cut off wavelength
 a ← core radius
 2.405 ← cut off normalized frequency for step index fiber $V_c = 2.405$

$$= \frac{2\pi \times 4 \times 10^{-6} \times 1.500 (0.06)^{1/2}}{2.405}$$

$$R_{cs} \approx \frac{20\lambda}{(n_1 - n_2)^{3/2}} \left(2.789 - \frac{0.996\lambda}{\lambda_c} \right)^{-3} = 1.214 \mu\text{m}$$

Critical curvature of single mode fiber -

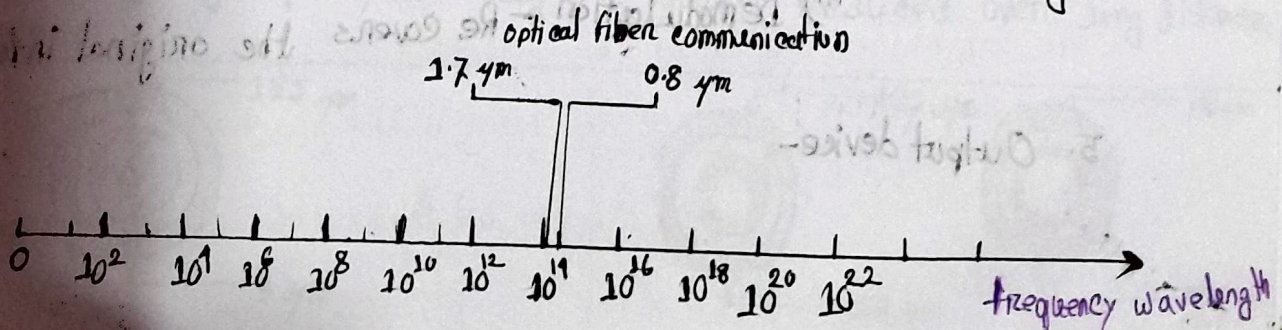
$$R_{cs} \approx \frac{20 \times 1.55 \times 10^{-6}}{(0.043)^{3/2}} \left(2.748 - \frac{0.996 \times 1.55 \times 10^{-6}}{1.214 \times 10^{-6}} \right)^{-3}$$

$$= 34 \text{ mm}$$

Previous year

Describe the electromagnetic spectrum showing the region used for optical fiber communication.

Optical fiber uses infrared light with wavelength between $0.8 \mu\text{m}$ and $1.7 \mu\text{m}$. This range is ideal for data transmission due to low signal loss. The visible spectrum is $(0.4 \mu\text{m} \text{ to } 0.7 \mu\text{m})$. Optical fiber communication operates just beyond the visible red light, in infrared region.



What are the difference between the overall communication and optical fiber communication?

Defination and diagram from 1st chapter page -

Explain optical fiber communication -

Diagram of APD detector and diagram of 1st page with explanation.

Typical fiber optic configuration with required schema -

APD detector diagram and explanation with →

1. Information source -

2. Transmitter -

i) Electric signal stage - convert information into electric signal

ii) Optical source - convert electric signal into light signal.

iii) Modulation - The light signal is modulated to carry the information.

3. Optical fiber -

i) Core

ii) Cladding

4. Receiver -

i) Optical detector - A photodiode or phototransistor that detected light signal

ii) Electrical signal stage - Process and amplifies

iii) Demodulation - Recovers the original information

5. Output device -

Advantage of optical fiber communication -

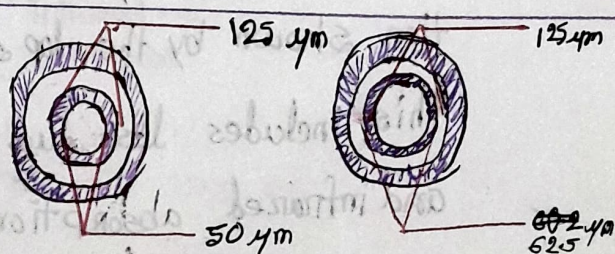
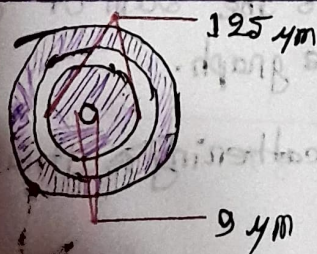
Page-3

Critical angle, Acceptance Angle, Refractive index

Page-3, 4

Difference between single mode and Multimode optic cable-

Single Mode fiber optic cable	Multimode fiber optic cable
1. A fiber optic cable with a small core that allows only one mode of light to pass through.	1. A fiber optic cable with a large core that allows multiple mode of light to pass through.
2. Core diameter - 9 microns	2. Core diameter - $50/62.5$ microns.
3. Cladding diameter - 125 microns	3. Cladding diameter - 125 microns
4. One mode of light	4. Multiple mode of light
5. Use for long distance communication	5. Use for short distance communication
6. Lower data capacity but better for long distance	6. Lower data capacity but limited for short distance
7. Maintain high signal quality over long distance	7. Quality reduced over long distance.



What is optical attenuation used for?

1. Evaluating transmission distance- Optical attenuation helps determine the maximum distance an optical signal can travel through.

2. Compare with metallic conductor- It allows for comparing fiber optics with metallic cable. Lower attenuation in fiber make them more suitable for long distance communication.

3. Signal Quality assesment- Attenuation measures the reduction in signal strength which helps assess and maintain quality.

4. System design and optimization- Knowing attenuation of a fiber optic system, we can ensure they are optimized for distance.

5. Loss Mechanism Analysis- Understanding attenuation helps identify and address various loss mechanism.

6. Performance monitoring- Regular attenuation measurements help monitor and maintain the performance of the fiber optic network.

Explain in details the intrinsic absorption loss mechanism with necessary illustration-

Optic losses in silica generally categorized into intrinsic and extrinsic type-

1. Total loss curve-

The total loss curve in the material is the sum of different attenuation shown by the top solid line in the graph.

This includes loss due to Rayleigh scattering, ultraviolet absorption and infrared absorption.

2. Rayleigh scattering -

- Rayleigh scattering cause attenuation due to imperfection and variation in the glass structure.

Rayleigh scattering curve decreases as the wavelength increases.

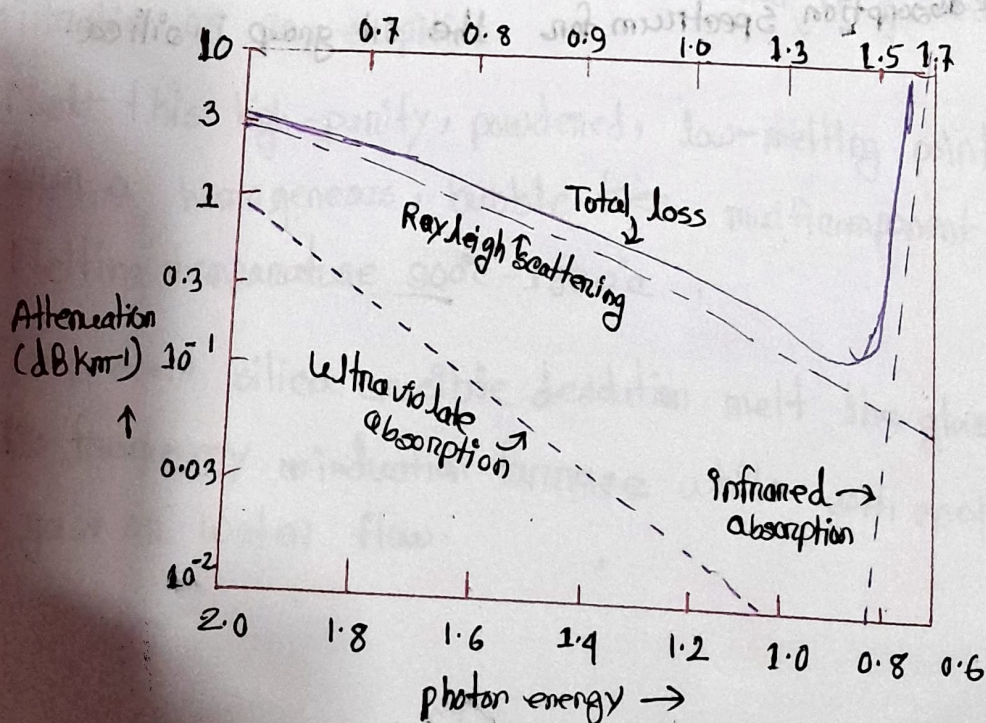
3. Ultraviolet absorption -

The dashed line indicates UV absorption, which increases at shortest wavelength.

The absorption occurs because the material absorbs higher-energy photons in the ultraviolet region.

4. Infrared Absorption -

At longer wavelength (beyond $1.6 \mu\text{m}$) infrared absorption becomes dominant. This due to vibrational modes in the glass structure that absorb longer wavelength.



Extrinsic absorption - 1. Metal impurities -

A major source of signal attenuation is extrinsic absorption from transition metal element impurities. Such as - Cr^{+3} , Cr^{+2} , Cu^{+2} , Fe^{+2} , Fe^{+3} , Ni^{+2} , Mn^{+2} , V^{+4} etc

Not primary

2. Hydroxyl group -

Another major extrinsic loss mechanism is caused by absorption due to water ^{Hydroxyl ion} dissolved in glasses.

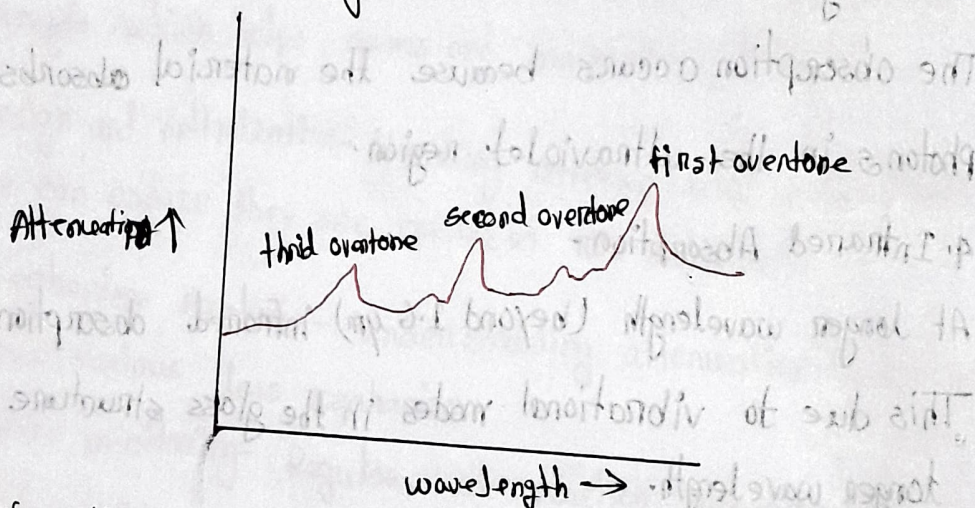


fig - the absorption spectrum for the OH group in silica.

